
GeoGenBench: Symbolically Verified Generation of Geometry Diagrams

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Abstract

Existing geometry benchmarks ask models to *interpret* a diagram; none ask whether a model can *produce* a correct diagram from a description. We introduce **GeoGen**, a methodology for automatically-verifiable benchmarks of generative geometric reasoning, and **GeoGenBench**, its first instance: 801 prompts paired with machine-checkable property lists. **GEOGEN** co-generates prompts and rubrics from shared Python source, then renders the model’s TikZ output and decides each predicate symbolically against compiled SymPy coordinates — no human grading, no vision-language judge. Benchmarking six frontier models under a raw-TikZ and a structured-IR strategy, we find the cost-quality Pareto frontier collapses to a single corner: three flagships tie within 1 p.p. at a $7\times$ cost spread under the IR, which lifts models by 2–25 p.p. (most for those weakest at raw TikZ). One vendor’s model rejects the IR schema outright — a portability gap, not a capability gap. A pre-registered $N=200$ human study finds moderate auto/expert agreement ($\kappa=0.455$) with strictly one-directional verifier lenience. We release the dataset, template engine, verification toolkit, and leaderboard.

1 Introduction

Frontier LLMs now author visual artifacts — slide decks, worksheets, technical illustrations. The most-deployed such artifact in K–12 education is the geometry diagram, yet every existing geometry benchmark evaluates the inverse direction: given a diagram, answer a question. None measures whether a model can *produce* a correct diagram from a textual description. **GEOGENBENCH** fills this gap.

Consider “Draw an acute triangle EFG with vertex angles 60° , 70° at E and F . Construct the altitude from G meeting EF at H .” A correct rendering must satisfy $\angle E = 60^\circ$, $\angle F = 70^\circ$, H on segment EF (not its extension), $\angle GHE = 90^\circ$, and acuteness. **GEOGENBENCH** encodes each clause as a machine-checkable predicate over the SymPy-compiled coordinates of the model’s rendered TikZ: in our pilot, one flagship placed H on line EF but *outside* the segment; another drew an obtuse triangle. The verifier flags each failure with a specific predicate verdict, not a holistic score.

We distinguish **GeoGen**, the methodology — (i) procedural co-generation of prompts and property-list rubrics from shared source, and (ii) a render-then-symbolic-verify pipeline driven by a declarative predicate language — from **GeoGenBench**, its 801-prompt plane-

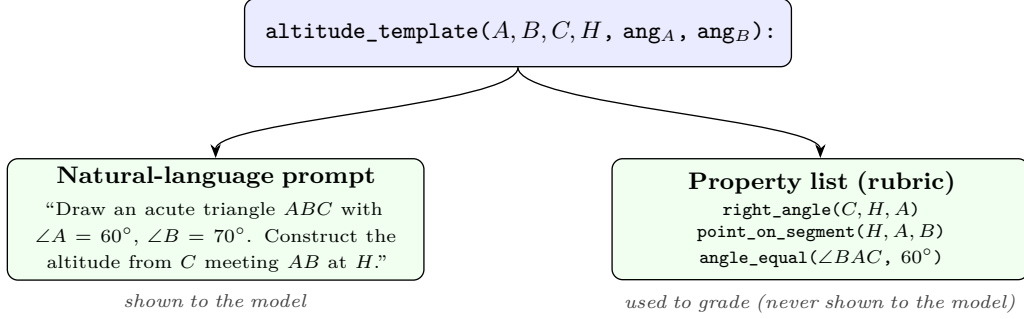


Figure 1: Procedural co-generation: one Python template emits both the prompt and the rubric, guaranteeing they stay mutually consistent.

33 geometry instance. Future GEOGEN-family benchmarks (coordinate-plane algebra, 3D solids,
34 engineering drawings) inherit the methodology and extend the per-domain primitives.

35 **Related work.** Geometry3K [6], GeoQA [3], UniGeo [4], MathVista [7], GeoEval [10], and
36 GeoBench [1] all measure diagram *interpretation*. On the generation side, T2I-CompBench++
37 [5] and GenExam [9] grade generated *images* via CLIP/MLLM or LLM judges, which cannot
38 tell whether a “right triangle” actually has a right angle — only whether the image looks like
39 one. GEOGENBENCH grades in symbolic space (TikZ $\rightarrow$ compiled SymPy geometry), where
40 a `right_angle` verdict is an exact numeric check, deterministic, reproducible, and $\sim 1000\times$
41 cheaper than an MLLM judge. Lean/Isabelle corpora and AlphaGeometry [8] machine-check
42 *proofs*; GEOGENBENCH extends automatic verification to *diagram generation*, occupying the
43 previously-empty generation-side, automatically-graded cell.

44 2 The GeoGen methodology and the GeoGenBench instance

45 **Pillar 1: Procedural co-generation.** Both the natural-language prompt and the
46 machine-checkable property list are emitted by the same Python template (Figure 1):
47 a single `altitude_template(A, B, C, H)` returns both the prose request and the predicate
48 list [`right_angle(A, H, B)`, `point_on_segment(H, B, C)`] any correct rendering must
49 satisfy. This eliminates rubric/question drift and provides a contamination defence: re-
50 viewers can re-roll the engine with fresh seeds at the same difficulty distribution. Where
51 co-generation is impractical, the methodology supports hand- or LLM-extracted prompts
52 with hand-authored rubrics, reported separately by source.

53 **Pillar 2: Render-then-symbolic-verify.** The model emits TikZ; a containerised LuaLa-
54 TeX renderer compiles it to SVG; an extractor compiles the same source to a coordinate-
55 bearing SymPy representation; a fixed-vocabulary, typed predicate language is decided against
56 that representation in microseconds per predicate. The v1 language comprises 17 symbolic
57 predicates (e.g. `right_angle`, `midpoint`, `collinear`, `parallel`, `tangent`, `angle_equal`,
58 `point_on_segment`) decided with numeric tolerance $\tau = 5 \times 10^{-3}$, plus 2 intentionally-
59 weaker source-string checks (label/mark presence) contributing only to a *loose* pass. The
60 language is declarative (authors specify *what* must hold, not *how* to check it), versioned, and
61 renderer-agnostic.

62 **Task and scoring.** A scenario is $S = \langle \text{prompt}, \mathcal{P}, \mathcal{L}, \mathcal{C} \rangle$; the model sees only the prompt
63 — properties are the rubric, never the input. Each output scores **pass** (all checks pass),
64 **soft_pass** (no failures, ≥ 1 check skipped), **fail**, or a generation/render failure.

65 **Composition.** 801 prompts: 600 templated scenarios from 30 procedural templates, 200
66 per difficulty tier — T1 (single shape), T2 (one auxiliary construction), T3 (multi-step
67 composition) — plus 201 curriculum scenarios LLM-extracted against the topical scaffold of
68 a state-adopted K–12 curriculum [2], adding breadth at the cost of LLM-authored ground
69 truth. Metrics are decomposed by source; the templated half serves as the calibration anchor.

70 3 Verification reliability

71 **Completeness and adversarial soundness.** A property list is *complete* if every diagram
 72 passing it is congruent (up to rigid motion and scale) to the intended construction. Auditing all
 73 30 templates yields 18/11/1 tight/loose/skeletal (63%/35%/2% of scenarios): loose templates
 74 pin the named relations but leave continuous parameters free, matching educational practice.
 75 For each of the 17 symbolic predicates we attempted adversarial TikZ that passes the
 76 predicate while contradicting the prompt; five predicates show soft spots, all addressable by
 77 hardening or template-side pairing, and we estimate $< 5\%$ of strict passes are vulnerable in
 78 the wild because templates pair check types so single-check loopholes close.

79 **Hardening.** Three verifier changes (closed-form intersection solving, a pgfmath-style
 80 coordinate evaluator, the `centroid` predicate) promoted 4.6% of pilot records from `soft_pass`
 81 to `pass` with zero demotions — and shrank the apparent structured-vs-raw strategy effect by
 82 roughly half, showing that part of the “strategy gap” was verifier weakness on raw outputs,
 83 not model weakness. All reported numbers are post-hardening.

84 **Human-correlation study.** A pre-registered study ($N=200$; stratified sample over the
 85 pilot; two domain-expert raters — a PhD mathematician and a K-12 mathematics educator
 86 — rating blind to the auto-verdict, then meeting to consensus) yields $\kappa = 0.455$ on the binary
 87 collapse (95% CI $[0.304, 0.590]$; balanced accuracy 0.684): **moderate agreement**, below
 88 the pre-registered 0.70 threshold. Inter-rater agreement is substantial ($\kappa = 0.704$), so the
 89 instrument is consistent; the gap is between raters and verifier. Crucially, the disagreement is
 90 *one-directional*: 18% of strict-pass records overturn to expert `fail`, zero the reverse. Per-tier,
 91 the disagreement concentrates entirely on T2 ($\kappa = 0.160$ vs $0.651/0.796$ on T1/T3), where
 92 property lists check the named auxiliary construction but not prompt-specified numeric
 93 parameters — the loose-completeness pattern made empirical (89% of disagreements code as
 94 “verifier missed prompt-numeric constraint”). Leaderboard rates are therefore conservative
 95 against the verifier-bias axis, and cell *differences* are largely preserved since the lenience
 96 applies uniformly across cells on affected templates; the v2 fix (numeric predicates lifting
 97 affected T2 templates to tight) is committed follow-up.

98 4 Experiments

99 **Setup.** Six production models across three vendors ($\sim 20\times$ price spread): Claude Opus 4.7,
 100 Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.5, Gemini 2.5 Pro, Gemini 2.5 Flash; temperature
 101 1.0; published April-2026 rates. Two reference strategies: **raw_code** (the model emits
 102 TikZ directly from the prompt plus a TikZ tutorial; zero scaffolding) and **structured** (the
 103 model emits typed JSON for a Pydantic intermediate representation — points, segments,
 104 intersections, circles, render ops, and *check* declarations — which code compiles to SymPy,
 105 then TikZ; declared checks run before render, so the strategy self-corrects with up to 3
 106 retries). The headline run covers all 12 (model $\times$ strategy) cells at 600 templated scenarios $\times$
 107 3 repeats; a subprocess-isolated runner with an OS-level kill timeout keeps per-cell coverage
 108 $\geq 91\%$ except where noted.

109 4.1 Findings

110 **Strategy choice helps unevenly.** structured lifts every model: GPT-5.5 +2.2 p.p.,
 111 Claude Haiku 4.5 +4.8, Claude Sonnet 4.6 +12.2, Claude Opus 4.7 +21.7, Gemini 2.5 Pro +24.7.
 112 The lift is largest for the models weakest at `raw_code` and smallest for the strongest.

113 **Frontier convergence.** Under structured, GPT-5.5 (94.1%, \$0.075/scen), Claude Sonnet 4.6
 114 (93.4%, \$0.088), and Claude Opus 4.7 (93.3%, \$0.532) span 0.8 p.p. at a $7\times$ cost ratio: the
 115 frontier-tier model carries no measurable quality premium under IR scaffolding, and the
 116 Pareto frontier collapses to the GPT-5.5 + structured corner. On `raw_code` the spread is
 117 wide (91.9% $\rightarrow$ 35.9%); Claude Opus 4.7 + `raw_code` is the most-dominated cell, beaten by
 118 GPT-5.5 + `raw_code` by 20 p.p. at $4.5\times$ lower cost. By tier, T1 is near-saturated for every
 119 Anthropic/OpenAI structured cell (v2 needs harder T1 templates); T2 is where strategy

Table 1: Headline leaderboard (600 scenarios $\times$ 3 repeats, post-hardening). Strict = **pass**; loose adds **soft_pass**. * marginally partial coverage; $\dagger$ blocked at the API layer (Section 4.1).

Model	Strategy	N	Strict	Loose	\$/scen.	Latency
GPT-5.5	structured	2,304*	94.1%	94.1%	\$0.075	33.6 s
Claude Sonnet 4.6	structured	2,861	93.4%	93.4%	\$0.088	19.0 s
Claude Opus 4.7	structured	2,292*	93.3%	93.3%	\$0.532	23.9 s
GPT-5.5	raw_code	1,800	91.9%	95.3%	\$0.058	14.8 s
Gemini 2.5 Pro	structured	1,140*	83.1%	83.1%	\$0.050	75.7 s
Claude Sonnet 4.6	raw_code	1,800	81.2%	95.3%	\$0.044	16.4 s
Claude Haiku 4.5	structured	2,232	75.3%	76.8%	\$0.032	8.0 s
Claude Opus 4.7	raw_code	1,800	71.6%	95.3%	\$0.262	13.8 s
Claude Haiku 4.5	raw_code	2,403	70.5%	76.0%	\$0.019	9.5 s
Gemini 2.5 Pro	raw_code	1,800	58.4%	89.9%	\$0.049	32.9 s
Gemini 2.5 Flash	raw_code	2,403	35.9%	42.2%	\$0.012	22.1 s
Gemini 2.5 Flash	structured	0 $\dagger$	—	—	—	—

first matters (Claude Opus 4.7 +43 p.p., Gemini 2.5 Pro +54 p.p.); T3 separates the frontier cells (GPT-5.5 93%, Claude Sonnet 4.6 87%, Claude Opus 4.7 82%). Most wrong outputs are *rendered diagrams with wrong geometry*, not broken LaTeX.

Portability: the schema-state cap. structured depends on schema-constrained generation, which OpenAI and Google compile to a server-side FSM with vendor-specific caps. Our IR schema ($\sim$ 20-variant discriminated union) compiles to thousands of states; Gemini 2.5 Flash rejects it at the API layer (HTTP 400: “*too many states for serving*”) before generating a single token. This is a portability gap, not a capability gap — and a caution for any benchmark shipping a multi-variant discriminated-union IR.

5 Limitations and conclusion

GEOGENBENCH measures whether a model can translate a textual specification of a 2D plane-geometry construction into a globally-consistent metric realisation; it does not test proof construction, 3D solids, or teaching utility. 35% of templated scenarios are intentionally loose (a pass means *a* diagram consistent with the stated constraints, not *the* canonical one), and the human study localises the resulting verifier lenience to T2 numeric constraints, with the v2 fix identified. On the cells run at headline N, the curriculum split is materially harder for every model measured (e.g. Claude Haiku 4.5 + **raw_code** drops 82.1% $\rightarrow$ 36.0%); headline findings are templated-split findings.

GEOGEN’s two pillars extend beyond plane geometry: co-generation reuses wherever a domain admits a Python-expressible construction, and render-then-symbolic-verify extends along the versioned predicate-language axis. In flight: a GenExam-Math integration [9] comparing symbolic verification against vision-LM judging on a shared 91-prompt subset, and coordinate-plane/3D extensions. We release the dataset, template engine, verification toolkit, Docker renderer, and leaderboard data under permissive licenses.

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